AGROCLOUD AI: CLOUD-NATIVE MULTILINGUAL SMART AGRICULTURAL ECOSYSTEM AND PRECISION AGRONOMY DECISION SUPPORT SYSTEM

A Major Project Report Submitted in Partial Fulfillment of the Requirements for the Degree of

BACHELOR OF TECHNOLOGY / BACHELOR OF ENGINEERING
IN COMPUTER SCIENCE & ENGINEERING / INFORMATION TECHNOLOGY

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CERTIFICATE OF ORIGINAL WORK

This is to certify that the project report entitled "AGROCLOUD AI: CLOUD-NATIVE MULTILINGUAL SMART AGRICULTURAL ECOSYSTEM AND PRECISION AGRONOMY DECISION SUPPORT SYSTEM" submitted by PUNITH is a bona fide record of work carried out under guidance and supervision in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science & Engineering.

The results embodied in this report have not been submitted to any other University or Institute for the award of any degree or diploma.

Project Internal Guide	**Head of Department**
Department of CSE	Department of CSE

CANDIDATE DECLARATION

I hereby declare that the project titled "AGROCLOUD AI: CLOUD-NATIVE MULTILINGUAL SMART AGRICULTURAL ECOSYSTEM AND PRECISION AGRONOMY DECISION SUPPORT SYSTEM" is an original piece of software engineering and research carried out by me. The software codebase, mathematical models, and technical documentation presented herein have been independently developed and verified.

Any material derived from academic publications, government open data portals (AGMARKNET, data.gov.in, Open-Meteo, Copernicus), or open-source software libraries has been properly acknowledged and cited.

Place: Bengaluru, Karnataka, India
Date: November 2026

Punith
Candidate & Lead Developer

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Lastly, I dedicate this project to the hardworking farmers of India whose resilience and dedication to feeding our nation inspire every line of code written in this system.

ABSTRACT

Indian agriculture sustains over 58% of the country's rural livelihoods, yet smallholder farmers continue to face catastrophic yield losses, severe climate risks, language barriers, and market asymmetry. Existing commercial agricultural applications operate in silos—offering either raw mandi prices without freight deductions, complex drone interfaces without actionable advice, or English-only mobile apps unsuitable for regional agrarian communities.

To overcome these systemic fragmentation gaps, this project presents AgroCloud AI, a unified, cloud-native, multilingual precision agriculture and decision support ecosystem. Built on Next.js 14, TypeScript, Tailwind CSS, PostgreSQL, and Supabase, AgroCloud AI integrates 14 production-grade modules into an accessible single-pane-of-glass dashboard localized natively in 12 major Indian languages.

Key technical innovations include a Realized Mandi Price Engine that deducts haulage distance freight, loading hamali, APMC market cess (1.5%), and weighbridge fees; a Multispectral Sentinel-2 Remote Sensing Engine computing calibrated NDVI and EVI vegetation vigor indices; a Cadastral Geospatial Parcel Mapping system utilizing the Spherical Shoelace algorithm; and a Computer Vision Crop Doctor providing dual organic and chemical treatment protocols.

Automated verification comprising 141 integration tests, strict zero-error TypeScript validation, and multi-tenant Row Level Security (RLS) across 30 PostgreSQL tables confirms that AgroCloud AI is a secure, production-ready, enterprise-grade solution poised to accelerate digital empowerment across Indian agriculture.

CHAPTER 1: INTRODUCTION

1.1 Background & Motivation

Agriculture serves as the foundation of the Indian economy, employing tens of millions across diverse agro-climatic zones. Despite remarkable advances in crop genomics and fertilizer technologies, smallholder and marginal farmers—who account for over 86% of total agricultural landholdings in India—face persistent structural bottlenecks:

- Severe Linguistic Exclusion: Most agritech applications are deployed exclusively in English or rudimentary Hindi, alienating millions of regional growers across Southern, Eastern, and Western states.
- Market Asymmetry & Intermediary Exploitation: Headline APMC mandi prices do not account for haulage distance, diesel freight, loading labor (hamali), market cess, or tare weighing, causing farmers to incur net losses when transporting produce to distant yards.
- Fragmented Agronomic Software: Farmers are forced to juggle separate apps for weather forecasting, pest detection, satellite vegetation indices, and farm expense ledgers.
- Hardware Barriers: Precision IoT soil monitoring hardware remains cost-prohibitive for smallholders without access to credit or digital infrastructure.

1.2 Overview of AgroCloud AI

AgroCloud AI is engineered as an all-inclusive, cloud-native agricultural software platform that unifies farm asset management, virtual IoT telemetry, satellite remote sensing, computer vision crop pathology, and logistics-aware mandi intelligence under a cohesive web dashboard available in 12 Indian languages.

CHAPTER 2: LITERATURE REVIEW

2.1 Digital Agriculture Evolution

Digital agriculture in India has evolved from first-generation SMS broadcasts (mKisan) to second-generation mobile applications (e-NAM, Kisan Suvidha). However, empirical studies reveal that fewer than 15% of farmers utilize e-NAM directly for spatial trade arbitrage due to a lack of integrated logistics calculations and real-time localized advisory.

2.2 Remote Sensing & Spectral Indices

Sentinel-2 Multi-Spectral Instrument (MSI) captures 13 spectral bands at 10-meter to 60-meter spatial resolution. The Normalized Difference Vegetation Index (NDVI) computes canopy vigor via (NIR - Red) / (NIR + Red). The Enhanced Vegetation Index (EVI) decouples atmospheric aerosol interference in high-biomass crops.

CHAPTER 3: PROBLEM DEFINITION & OBJECTIVES

3.1 Problem Definition

Smallholder agriculturalists in India operate in an information-opaque environment characterized by linguistic exclusion, high hardware capital barriers, volatile input costs, and deceptive wholesale market pricing. The central technical challenge is to design, implement, and verify an integrated cloud-native web platform that eliminates intermediaries, democratizes satellite and IoT telemetry, and provides actionable agronomic decisions natively in the farmer's mother tongue.

3.2 Project Objectives

• Objective 1: Design a high-performance web application supporting 12 Indian languages with sub-second page transitions.

• Objective 2: Engineer a 30-table relational schema on PostgreSQL with strict Row Level Security (RLS) guaranteeing complete tenant isolation.

• Objective 3: Implement mathematical models for satellite remote sensing (NDVI/EVI) and cadastral spherical shoelace acreage calculation.

• Objective 4: Build a Realized Mandi Logistics Calculator deducting diesel freight, labor hamali, APMC cess (1.5%), and weighbridge fees.

• Objective 5: Develop a Virtual IoT Environmental Telemetry Simulator based on FAO-56 Penman-Monteith crop water models.

• Objective 6: Integrate a multimodal AI Crop Doctor providing dual chemical and organic treatment protocols from leaf photos.

• Objective 7: Validate the complete software platform with 100% automated test coverage and zero TypeScript compile errors.

CHAPTER 4: METHODOLOGY & MATHEMATICAL MODELS

4.1 Spherical Shoelace Geodesic Acreage Algorithm

To compute true ground acreage from GPS coordinates drawn by the farmer on an interactive Leaflet/OSM map, standard Cartesian geometry introduces intolerable distortions due to the Earth's curvature. AgroCloud AI implements the Spherical Shoelace algorithm on the WGS-84 authalic sphere (radius R = 6,371,007 meters).

Area (sq meters) = (R^2 / 2) * |Sum (lambda_{i+1} - lambda_i) * (2 + sin phi_i + sin phi_{i+1})|

Ground Acreage = Area (sq meters) / 4046.8564224

4.2 Mandi Freight & Realized Revenue Model

Net Realized Price per Quintal = (Gross APMC Revenue - (Freight Cost + Hamali Labor + 1.5% APMC Cess + Weighbridge Fee)) / Total Quintals

CHAPTER 5: SYSTEM ARCHITECTURE & DATABASE DESIGN

5.1 Multi-Tier Cloud Architecture

AgroCloud AI employs a modern 3-tier cloud architecture:

• Tier 1: Next.js 14 App Router presentation layer with Tailwind CSS, Recharts, and 12-language i18n engine.

• Tier 2: TypeScript computation service layer managing geospatial mathematics, satellite spectral processing, and agronomic rule evaluation.

• Tier 3: Supabase PostgreSQL 15 database featuring 30 relational tables, RLS tenant isolation, and private storage buckets.

5.2 PostgreSQL 30-Table Relational Schema

The database encompasses 30 tables covering user profiles, preferences, farms, fields, crops, farming activities, expenses, IoT sensors, telemetry readings, monitoring alerts, AI conversations, crop diagnoses, APMC mandi records, transport calculations, farm geometries, satellite observations, drone surveys, and simulation scenarios.

CHAPTER 6: IMPLEMENTATION DETAILS

6.1 Modern Full-Stack Stack

The platform is built with Next.js 14, TypeScript 5.7, Tailwind CSS, PostgreSQL, and Supabase. Over 120 standardized translation keys are localized across 12 Indian languages.

6.2 Core Functional Modules

Implemented 14 distinct modules including Executive Dashboard, Farm & Crop Management, Virtual IoT Telemetry, Weather & Evapotranspiration Advisories, AI Agronomist, Crop Doctor Leaf Pathology, Mandi Intelligence, Digital Farm Twin, Sentinel-2 Satellite Analytics, Drone Survey Analytics, What-If Farm Simulator, and Cadastral Parcel Mapping.

CHAPTER 7: RESULTS AND PERFORMANCE EVALUATION

7.1 Automated Testing Results

The platform achieved 100% test pass rate across 141 integration tests in scripts/test-phase7.mjs:

- Suite 1: 12-Language Localization Integrity: 96/96 Passed
- Suite 2: Satellite Spectral Remote Sensing Mathematics: 4/4 Passed
- Suite 3: Geospatial Geodesic Area Calculation: 2/2 Passed
- Suite 4: Mandi Logistics & Realized Pricing Engine: 3/3 Passed
- Suite 5: What-If Farm Agronomic Simulator: 2/2 Passed
- Suite 6: Drone Survey Upload & Size Validation: 3/3 Passed
- Suite 7: Supabase Master Database Schema Verification: 26/26 Passed
- Suite 8: Security & Secret Leak Prevention: 5/5 Passed

TypeScript Type Check: Exited with code 0 and ZERO errors.

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Summary of Achievements

AgroCloud AI demonstrates that cloud-native architectures, satellite remote sensing, and localized decision intelligence can be effectively unified into an accessible, low-bandwidth web platform that empowers Indian farmers with data-driven autonomy.

8.2 Future Scope

- Edge Drone Real-Time Video Streaming over WebRTC.
- Offline Progressive Web App (PWA) with IndexedDB client caching.
- Blockchain supply chain traceability for organic crop exports.

REFERENCES & APPENDIX

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Appendix

Complete source code, schema migrations, and documentation files are maintained in the project repository root.